

Hello World!

Welcome to PSY 210



Methods in Psychological Research



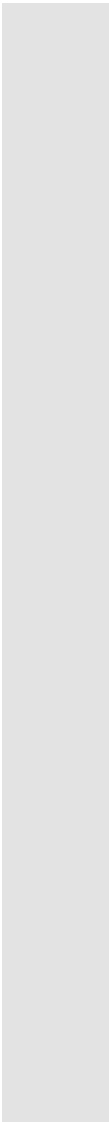
210 is an introduction to methods in psychological research,



with an emphasis on

statistical thinking,
best practices, and
practical skills

What is this
class?



Before I get too far
ahead of myself ...

Lecture Roadmap

All lectures will start with a roadmap, including today:

- Overview of the Course
- Learning Outcomes
- Knowledge is Power
- What is Statistics?

The roadmap gives us structure as I lecture

- Trust Me, you don't want me to lecture without one

Overview

Course

PSY 210:
Methods
Officially Face2Face

Professor

Prof. Mason Garrison
Greene 438
Office Hours:
by appointment
[https://calendly.com/
smasongarrison/](https://calendly.com/smasongarrison/)

Assistant

Tukey

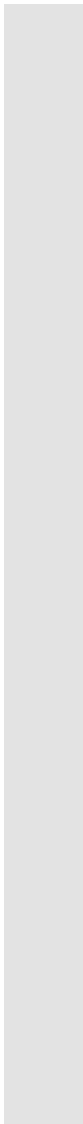


- Call Me... Prof. Mason (preferred)
 - Prof. Garrison, Dr. Mason, or Dr. Garrison are also ok
 - Not Sarah, Mason, Mrs. Garrison, Ms. Garrison

Introductions

About me

- Born in Upstate New York, Undergrad in St. Louis, Grad School in Nashville, and Professing in Winston-Salem
- Quantitative Psychology and Behavior Genetics:
 - Applied to Understanding the Relationship between Health, Wealth, and Individual Differences.
- I love stats and cats



Learning Goals

How to Think About Them

Ways for teachers to define what we want students to be able to do.

Used in assessments (e.g., quizzes and papers)

I will be asking you to demonstrate that you have achieved these outcomes.

You can also try to evaluate whether you've achieved the learning outcomes.

Practice making questions that will test your understanding, share them with others, and try each other's questions.

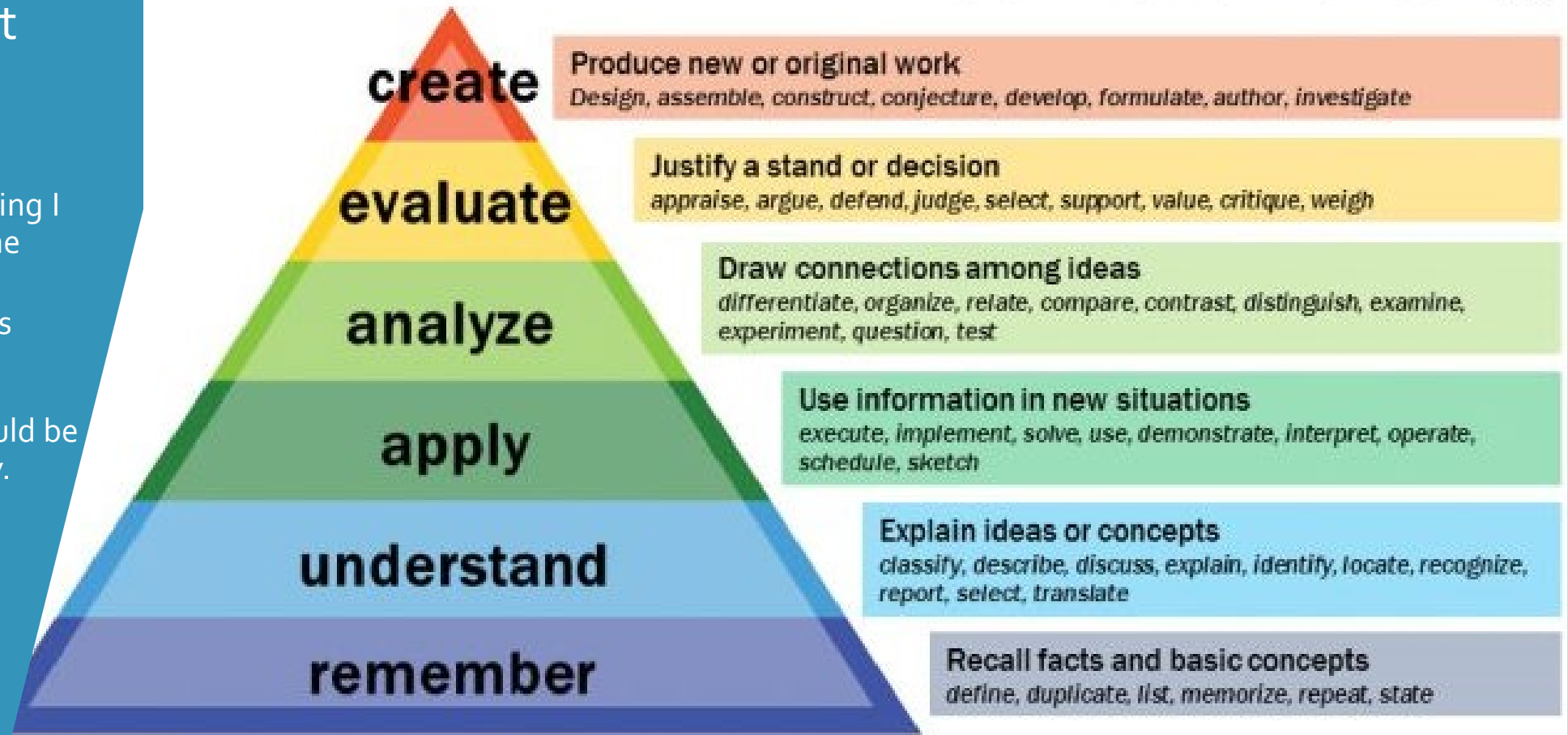
How to Think About Them

Note the verb!

The depth of understanding I expect is articulated in the depth of the verb, and is based on Bloom's Taxonomy.

Your skills in an area should be moving up the taxonomy.

Bloom's Taxonomy



Learning Goals: Course Level



Identify research claims in both popular and scientific sources and classify them as frequency, association, or causal claims.



Separate the claim from the study:

Does the data support the claim?



Evaluate the extent to which researchers have measured their variables reliably and with validity.



Discriminate between experimental, quasi-experimental, and correlational studies.



Explain why correlational studies cannot support causal claims.



Explain when and why an experiment can support a causal claim.



Explain when we can generalize from a study and explain when generalization is not a priority.



Describe the ethical guidelines that apply to research in psychology.



Become critical thinkers and consumers of research in the academic realm and popular media.

Learning Goals: Course Level

Course Milestones

Three (3) Exams

- Friday, 10/03
- Friday, 11/14
- During Finals, 12/13

Two (2) Short Critiques

- Friday, 10/17
- Friday, 12/05

One (1) Reading Quiz

- Per module

Two (2) Engagement Activities

- per week/module

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You can drop
some . . .

Lowest Drops



Exams:

The best 2 out of 3 exam grades will be used to determine your total exam grade,

- As long as you **pass the final exam.**



Reading Quizzes

The best 12 out of 14 quizzes will be used to determine your total quiz grade



Short Assignments:

The best 1 out of 2 will be used.

Resources

Textbook:

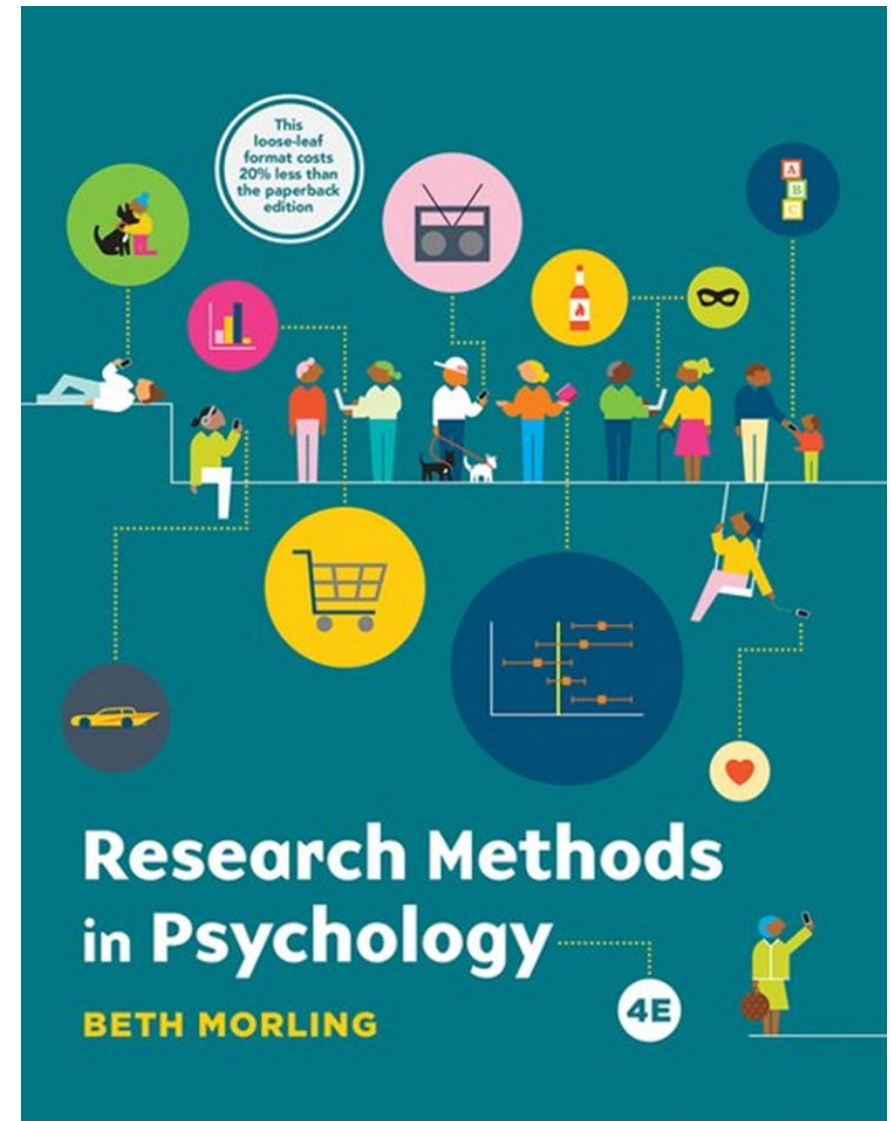
- B. Morling. Research Methods in Psychology. 5th Edition. (4th is probably fine)

Course Notes:

- <https://psychmethods.github.io/coursenotes/>

Canvas:

- <https://wakeforest.instructure.com/courses/66374>



Exams

Format

- Mostly Short Answer Questions
- On Canvas

Content

- Based on the Learning Goals
- I **randomly** select from the study guide to make the questions
- Each question has a scoring rubric.
 - (If you want to see your quiz responses after... you have to schedule a meeting with me)

Open Resource

- The exams are open-note, video, internet.
 - HOWEVER: they are not open to other people
- Everything must be written in your OWN words.
- Use the learning goals to organize your notes

1

Critique of methods-related media coverage.

2

Critique of methods-related video (e.g, YouTube).

Short Assignments ($\approx$ 2.5 pages)

Two (2) Engagement Activities

- Low stakes way to maintain a sense of community
 - These will be graded pass/fail
 - And can be completed a lot of different ways
 - including discussion boards,
 - in-class activities,
 - reflections
 - reaction videos
 - sharing a cool example you've found
 - (I'm very open to suggestions for these!)
- You must submit them by Friday at 6 pm!





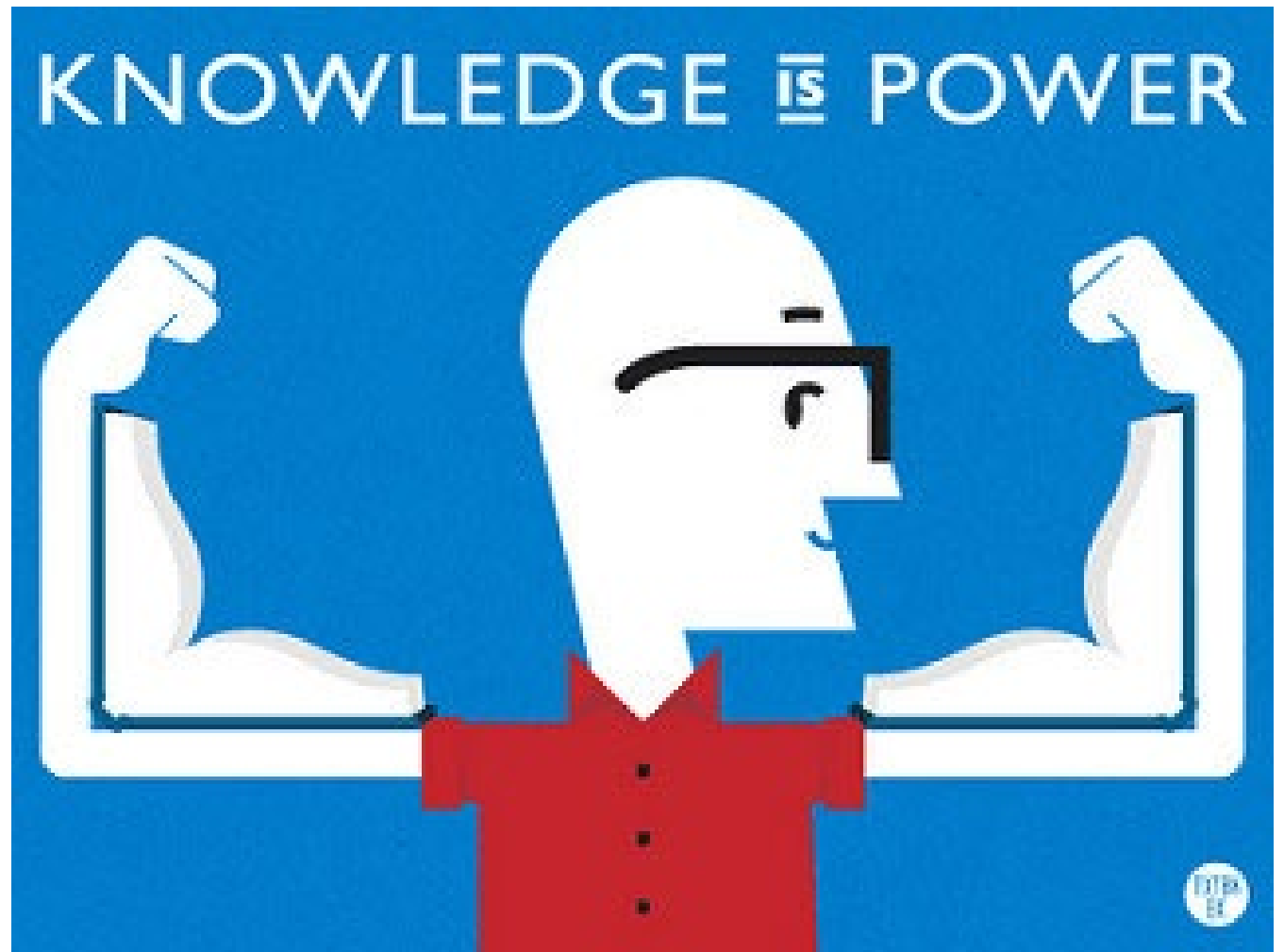
In-Person Course Content

- I will give live lectures that correspond to the module materials.
 - Typically, this will be on Mondays and Wednesdays.
- I will provide in-class activities
 - (discussion, demonstrations, review sessions)
 - that count as engagement activities.
 - Typically, this will be on Fridays.

Wrapping
Up...



Knowledge
is Power



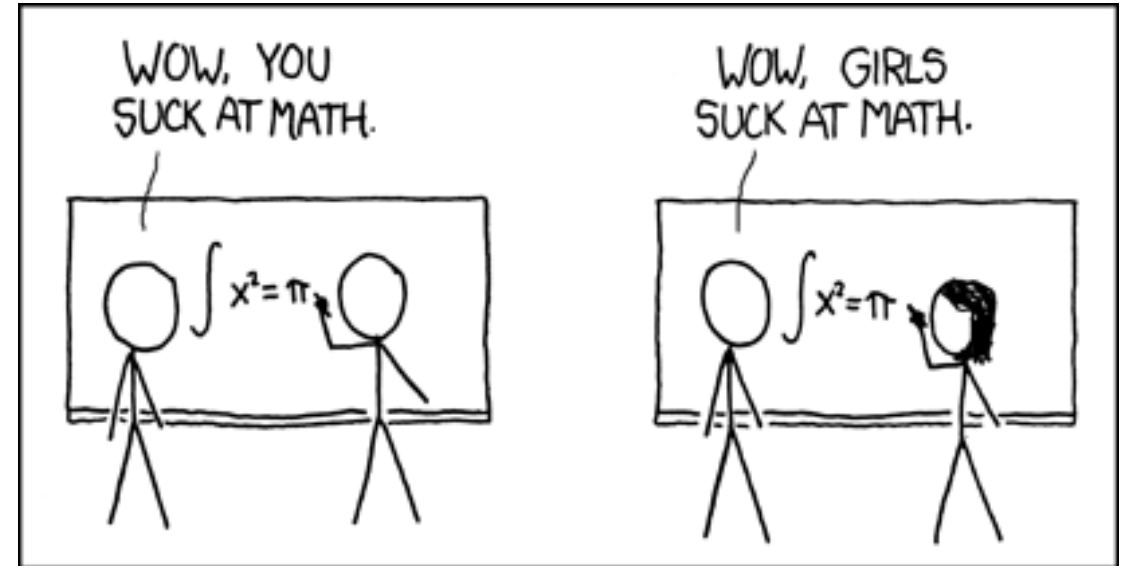
Knowledge is Power

- Stereotypes exist about how good certain people are at
 - Science,
 - Math, and
 - School.
- Or how some people are naturally smart.

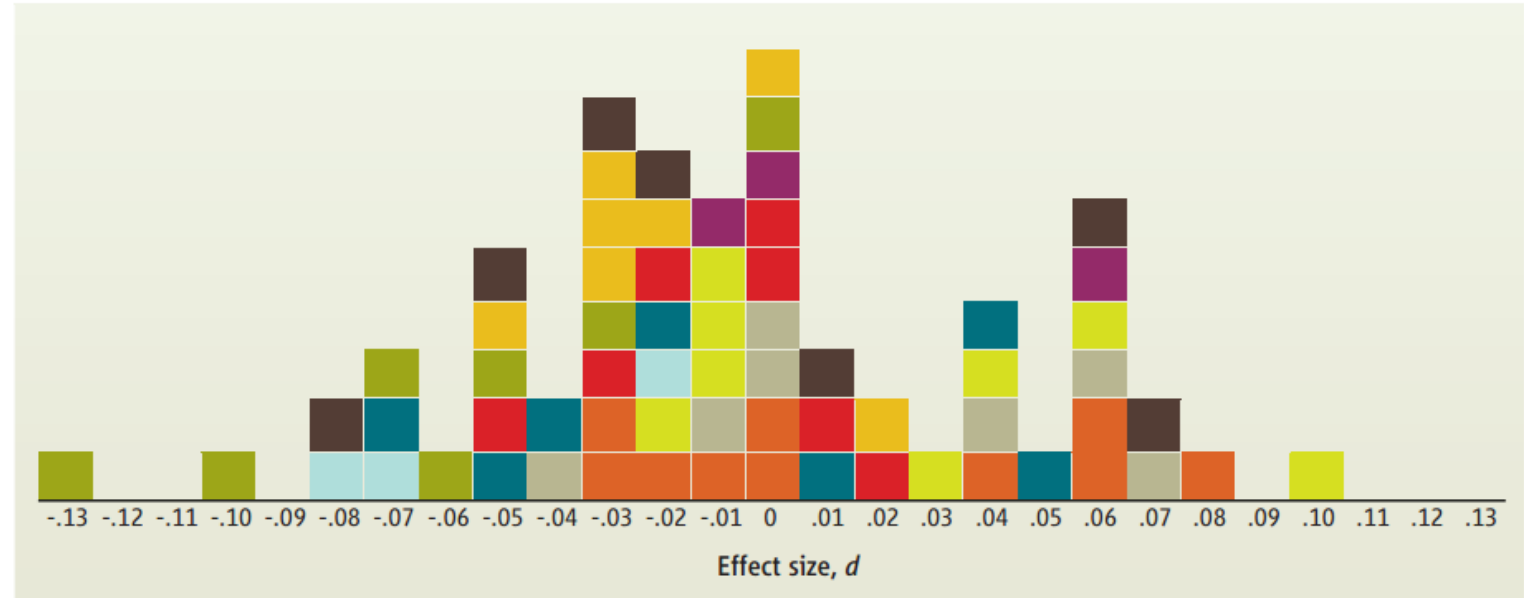
Knowledge is Power

These stereotypes form at an early age (Cvencek, Meltzoff, and Greenwald 2011).

For example, some people think that girls are worse at math than boys.



Actually,
boys and
girls perform
the same on
math
achievement
tests (Hyde,
et al, 2008).



Effect sizes across grades and U.S. states. The weighted mean is 0.0065, consistent with no gender difference. Each square represents the effect size for one grade within one state. New Mexico (pea green), Kentucky (pink), Wyoming (dark brown), Minnesota (teal), Missouri (red), West Virginia (gold), Connecticut (tan), California (orange), Indiana (yellow), New Jersey (purple).

Hyde, J. S., Lindberg, S. M., Linn, M. C., Ellis, A. B., & Williams, C. C. (2008). Gender similarities characterize math performance. *Science*, 321(5888), 494-495.

Knowledge is Power

- However, stereotypes about math performance can undermine how well you do on a test, by
 - Increasing test anxiety,
 - Discouraging you from seeking out math activities, and
 - Giving up sooner.
- One of the best ways to protect yourself from “stereotype threat” is to be aware of it. (Johns, Schmader, & Martens, 2005)
- Johns, M., Schmader, T., & Martens, A. (2005). Knowing is half the battle, teaching stereotype threat as a means of improving women's math performance. *Psychological Science*, 16(3), 175-179.

Knowledge is Power:
Activity!



Knowledge is Power: Activity!

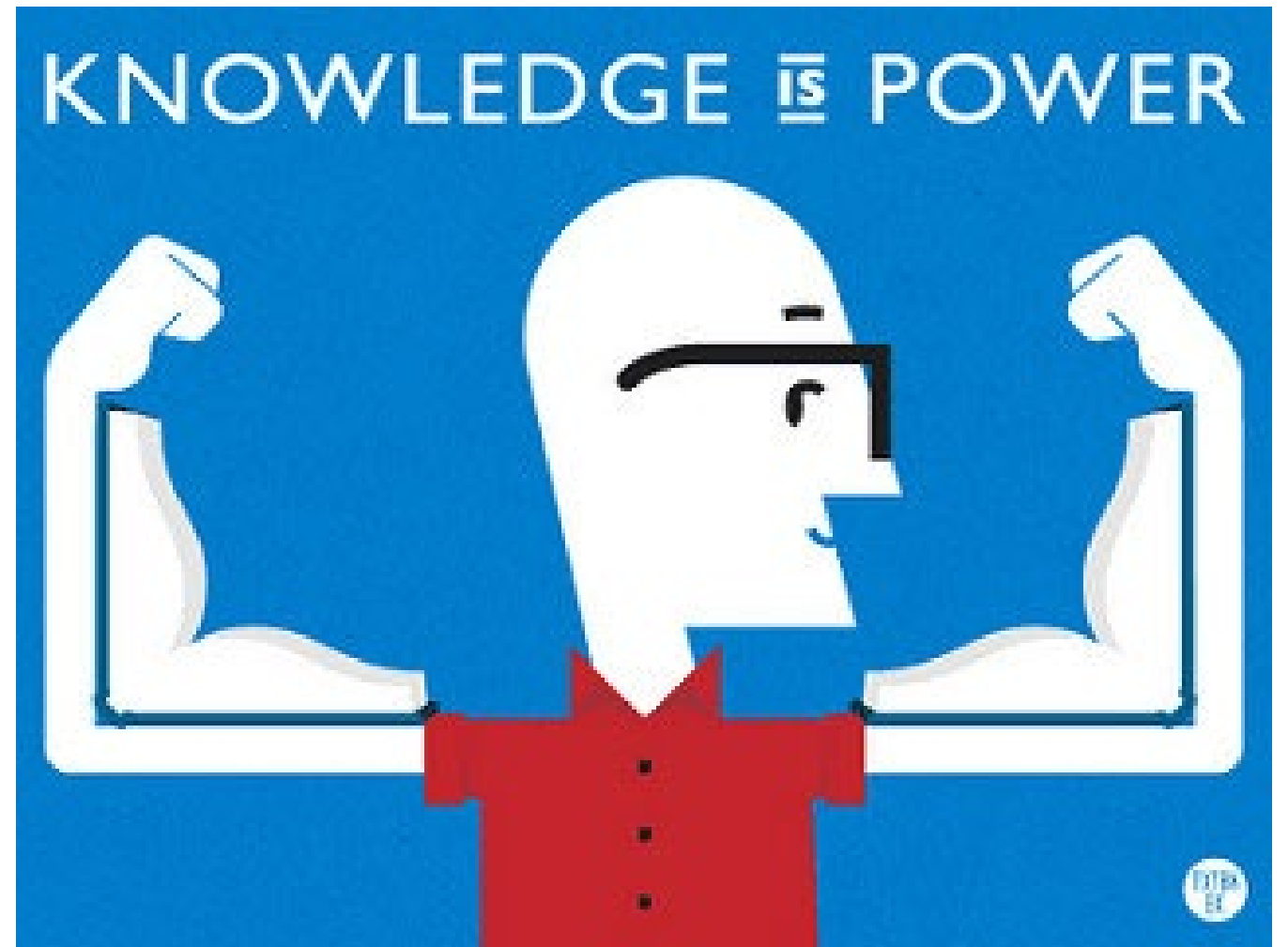
Write

1. specific concerns of the course
 2. their favorite class
- their dream job (now and when you were in kindergarten).

Then we'll introduce ourselves and share about our favorite and current dream job



Wrapping
Up...
Knowledge
is Power



What is Statistics?

What is Statistics?

Statistics is the methodology for

- *collecting,*
- *analyzing,*
- *interpreting and*
- *drawing conclusions from information.*

Scientists and mathematicians have developed these methods for

- *interpreting and drawing conclusions*
- *from collected data.*

What is Statistics?

- Everything remotely associated with the collection, processing, interpretation and presentation of data
 - belongs to the domain of statistics, including
 - *detailed planning of that precedes all these activities.*

What is Statistics?

- Statistician : Mathematics ::
 - *Actuary : Insurance*
 - *Biostatistician : Biology*
 - *Chemometrician : Chemistry*
 - *Econometrician : Economics*
 - *Psychometrician : Psychology*
 - *Quantitative psychologist : Psychology*

What is Statistics?

Statistical methods can be used to find answers to questions, like

- *What kind and how much data need to be collected?*
- *How should we organize and summarize the data?*
- *How can we analyze the data and draw conclusions from it?*
- *How can we assess the strength of the conclusions and evaluate their uncertainty?*

That is, statistics provides methods for

- *Design: Planning and carrying out research studies.*
- *Description: Summarizing and exploring data.*
- *Inference: Making predictions and generalizing about phenomena represented by the data.*



Wrapping Up...